Calories Burnt Prediction Using Machine Learning

Raghad Oteef
College of Engineering and Computer Science-Computer Science
Jazan University

Aryam Nasser
College of Engineering and Computer Science-Computer Science
Jazan University

Wejdan Hakami
College of Engineering and Computer Science-Computer Science
Jazan University

***Abstract*—Knowing the number of calories burned is a difficult problem we face in our lives today due to the prevalence of obesity, unlike in previous centuries. Machine learning helps us train a model to predict calories burned based on a number of features, thus solving the problem of calculating calories burned. The project includes finding the dataset and exploring it in general, and performing Data Preprocessing, building the model using two different algorithms, training it, and testing it in order to find the best algorithm to build a more accurate model. We used this dataset (predicted calories burned) to train our model, and compared the linear regression and decision tree algorithms. We concluded that the linear regression algorithm yields better results as it is more accurate. In our model, the best achievement was that our model accurately predicted the calories burned, which helps us in our lives in general. This shows the importance of our project, as it solves a major problem facing society and leads us to a more conscious society regarding the calories they burn.**

Keywords—Machine Learning, Calories Burnt Prediction, Regression Algorithms.

I. INTRODUCTION (*HEADING 1*)

In recent years, obesity rates have risen due to decreased physical activity and the prevalence of high-calorie foods. According to the 2024 statistics from the General Authority for Statistics in Saudi Arabia, the obesity rate among population aged 15 and above is 23.1%, while 45.1% of individuals in this age group are classified as overweight. Additionally, the prevalence of obesity among children aged 2 to 14 years stands at 14.6%, with 33.3% of children in this age range being overweight [1]. Calculating calories burned helps in weight management and choosing appropriate activities and exercise durations.

Benefits of Using Machine Learning in Predicting Calories Burned:

Accuracy and Insights: Machine learning delivers remarkable accuracy by identifying patterns in complex datasets. Unlike traditional approaches, which are limited by static algorithms, machine learning models continuously adapt to new data, refining their outputs over time [2]. In our problem, when new data is added, machine learning can utilize it and improve its accuracy, unlike static algorithms.

Predictive Capabilities for Better Planning and Efficiency[2].

Machine learning models provide predictive capabilities for calorie prediction, which helps users better plan the duration of physical activities based on heart rate and other factors, thus improving efficiency in several aspects (such as exercise duration and type).

In our project, we selected a dataset (calories Burnt) and prepared it by exploring, cleaning, and displaying the data in graphs and creating descriptive statistics. We then compared two algorithms (Decision Tree and Linear Regression) to find the best algorithm for a model to better predict calories burned.

II. METHODOLOGU

A. Dataset Description

In this project, we used a dataset titled (Calories Burnt Prediction) to predict the number of calories a person burned during exercise based on some biological measurements.

The dataset consists of 15,000 row and 9 columns, where 8 columns represent features and one column for the target variable. The columns are User_ID, Gender, Age, Height, Weight, Duration, Heart_rate, Body_Temp. We will learn more about the columns in greater depth and detail through the table and graphs below.

This dataset was selected because it provides diverse and accurate information on calories burned during exercise. The goal is to build a model that can predict the number of calories a person burns while exercising. This topic is also of interest to all segments of society, from adults to young people, as obesity rates have increased recently, and we no longer have enough time to focus on these details. Therefore, we need something to help us calculate calories burned, motivating us to exercise and informing us of the number of calories burned during each workout.

Dataset Link:

https://www.kaggle.com/datasets/ruchikakumbhar/calories-burnt-prediction

Table 1 : Describe the dataset features

Feature name	Describtion
1.User_ID.	User_ID is the unique number for each person The data type is (integer).
2. Gender:	It specifies whether the person is male or female. The data type is (string) , and it is of values is two.
3. Age:	Age is the age of each person in the database . The data type is (integer).
4. Height:	A person's height is measured in centimeters. The data type is decimal.
5. Weight:	A person's weight is specified in kilograms. The data type is decimal.
6. Duration:	This is the duration of the exercise performed by the person. The data type is decimal.

7.Heart_rate:	Heart rate during exercise. The data type is decimal.
8. Body_Temp:	Body temperature during exercise. The data type is decimal.

Target Variable	Describtion
9.Calories.	Calories is the target data column for prediction and contains the number of calories burned during exercise. The data type is decimals.

B. Data Preprocessing

1) **Load the dataset into Python:**
We Load the Calories Burnt dataset into Python using Panda Library.

Pandas Library is a Python library rich in tools for working with datasets and provides a wealth of rich data structures. It is used in a variety of fields, including statistics and data analysis. The name Pandas is derived from "Panel Data".[3]

There are two main types of data structures in the Pandas library: series and data frames. A Data Frame is a data structure similar to a spreadsheet, consisting of an assortment of arranged sections. Each section can contain values of different types, such as numeric, strings, or other formats. It can also be defined as a Python object containing rows and columns. [4]
Pandas provides a method that allows us to extract data from a CSV file and then create a data frame. [4] And that is done through the read_csv method which takes the path of the CSV file as an argument then we store the result in a Data Frame variable.

2) **Data exploration:**
Data exploration means that we illustrate the overall pattern of the data and extract it so that we have an idea about the type of data and its cleanliness, even if we do not know what we are looking for[5].
Here, we explored the data by first displaying the first few rows using (df.head()), then we showed the data types of the columns as indicated in the table above using (df.dtypes), which were as follows: (User_ID = int64, Gender = object, Age = int64, Height = float64, Weight = float64, Duration = float64, Heart_Rate = float64, Body_Temp = float64, Calories = float64).
Secondly, there are some datasets that often contain certain stress points, such as missing data or duplicate data [6].so we searched for missing values using (df.isna().sum()), and the result was that we found no missing values. After that, we checked for duplicate values using (df.duplicated().sum()), and again, we found no duplicated values.
In summary, when we explored the data, we found that it contains 15,000 rows and 9 columns and does not include any duplicate or missing values; in other words, we find that the dataset it already clean.

3) **Descriptive statistics:**

Generate descriptive statistics and find the arithmetic mean, median, standard deviation, count.

Mean:
The most famous and widely used measure of central tendency is the arithmetic mean. We can use it with both types of data, whether continuous or discrete, and it is often used with continuous data. But it has a fundamental drawback, which is that it is very sensitive to the influence of outlier values, which are values that are much larger or much smaller than the rest of the values [7]. The arithmetic mean can be calculated by adding all the values and dividing by the number of values.

	0
User_ID	1.497736e+07
Age	4.278980e+01
Height	1.744651e+02
Weight	7.496687e+01
Duration	1.553060e+01
Heart_Rate	9.551853e+01
Body_Temp	4.002545e+01
Calories	8.953953e+01

Median:
The median is the number that is in the middle of the data sorted by size. It is less affected by outlier values than the arithmetic mean. When you have an even number of data points and there are two numbers in the middle of the ordered data, we take the two middle numbers and divide them by 2 to get the median.[7]

	0
User_ID	14997285.0
Age	39.0
Height	175.0
Weight	74.0
Duration	16.0
Heart_Rate	96.0
Body_Temp	40.2
Calories	79.0

Standard Deviation:
Gives an idea of how close the entire set of data is to the average value. Data sets with a small standard deviation have tightly grouped, precise data. Data sets with large standard deviations have data spread out over a wide range of values.[8]

	0
User_ID	2.872851e+06
Age	1.698026e+01
Height	1.425811e+01
Weight	1.503566e+01
Duration	8.319203e+00
Heart_Rate	9.583328e+00
Body_Temp	7.792299e-01
Calories	6.245698e+01

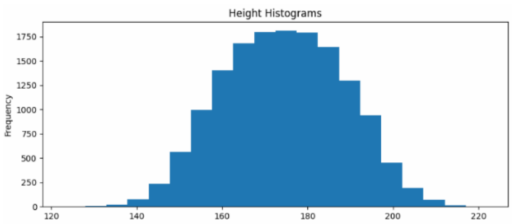
Count:
The number of rows in each column. As we observed from exploring our data, rows, and columns, we found that each column contains 15,000 row values in our data.

	0
User_ID	15000
Gender	15000
Age	15000
Height	15000
Weight	15000
Duration	15000
Heart_Rate	15000
Body_Temp	15000
Calories	15000

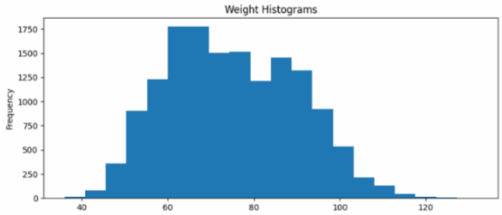
4) Create visualizations:

Histograms for feature distribution:

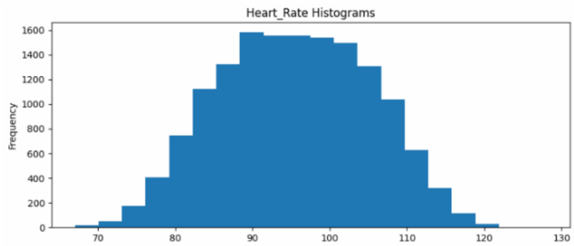
A histogram is a type of graphical representation, similar to a bar chart, but it is used for numerical data. It is used in statistics to show the distribution of numerical data. A histogram is designed for continuous data because it groups it into a logical range known as "bins". Histograms help us visualize the distribution of data, making the data clear and highlighting trends and patterns.[9]



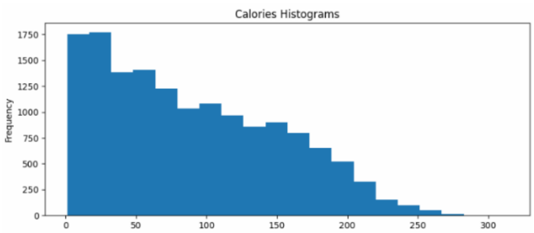
According to the histogram, In the x-axis, we set the height, and in the y-axis, we set the frequency, which is the number of people in each height. We noticed that the height ranged approximately from 130 to 210, The most frequently occurring height in the dataset ranges between 170 and 180 approximately, which is 1750 people. We also noticed there are fewer people with heights between (130 and 140) and also between (205 and 210), which are very few in the dataset.



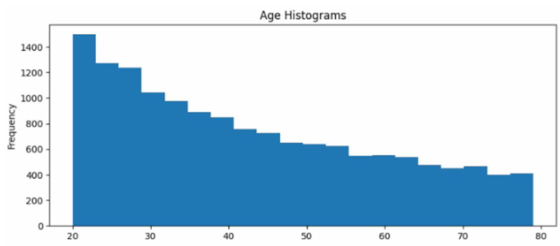
According to the histogram, in x-axis, we specify the weight, and in y-axis, we specify the frequency, which is the number of people in each weight category. We observed that the weight range is between 30 and 130 kg, and that the 60-70 kg weight range is the largest, with approximately 1750 people in the dataset. This might indicate that this group tends to maintain their health. We also noticed a small number of people weighing between 30 and 40 kg, which might make sense since they don't need to count calories burned. However, what's surprising is that the 120-130 kg weight range is also small, even though they are supposedly the group most concerned with maintaining their health and need to know their calorie burn.



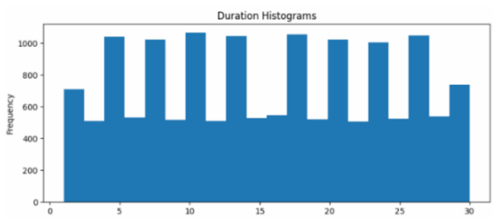
According to the histogram, the heart rate of most people ranges approximately between 88 and 92, while normal heart rates range between 60 and 100. Heart rates can increase during physical exertion when exercising (exceeding 100 beats) or even with slight effort. Therefore, we conclude that exertion raises the heart rate, which improves cardiovascular fitness and the rate of calorie burning, even if only slightly. We also observe in our community that the people with heart rate between 70 and 80 below the normal range, which may be due to weak heart muscles, mostly due to not exercising regularly or exerting less effort than normal.



According to the histogram, the x-axis represents the calories burned during exercise, while the y-axis represents the number of frequencies —that is, the number of people who burned the same number of calories. The calorie values generally range from approximately 0 to 280 calories. We can observe that the highest number of frequencies is between 0 and 33 with approximately 1750, while the lowest is between 250 and 280. We can also see that the higher the number of frequencies, the lower the number of people having those calories. Therefore, we can conclude that most people in our data tend to burn fewer calories.

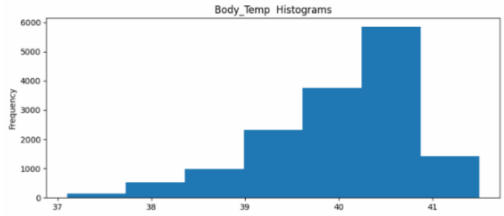


According to the histogram, in x-axis, we specify the age, and in y-axis, we specify the frequency, which is the number of people in each age group. We observed that the age groups are between 20 and 78, and we noted that the average age is between 20 and 25. This might make sense, as they are younger and have sufficient awareness that makes them want to know their calories burned, maintain their physical health, and improve their appearance. However, the age group between 70 and 78 is not very small, but it is the smallest in the dataset. This might be due to a lack of awareness about the importance of health and knowing the number of calories burned.



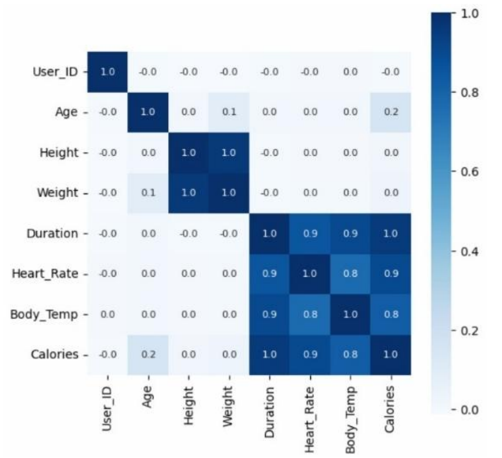
According to the histogram, In the x-axis, we selected Duration, which is the time a person spends exercising and, in the y-axis, we selected Frequency, which is the number of people they have the same duration. We noticed in the chart that the duration of exercise varied between different time periods for different individuals. This may be due to differences in individuals' endurance, as older adults may have shorter durations than younger people who are at the peak of their strength and youth, which leads to their inability to endure long periods of exercise.

However, this may not always be true, which is why the durations vary.



According to the histogram, in the x-axis, we specify body temperature, and in the y-axis, we specify the frequency, which is the temperature that people reached during exercise. The graph shows us that the temperatures are between 37 and 42, and the average temperature is between 40 and 41. This may be due to the physical effort they exert during exercise. Thus, we can know that during exercise, the temperatures rise, and the lowest temperature is between 37 and 38. This may be because the exercise was light and they did not exert much physical activity, for this reason the temperatures were low.

Correlation heatmap:

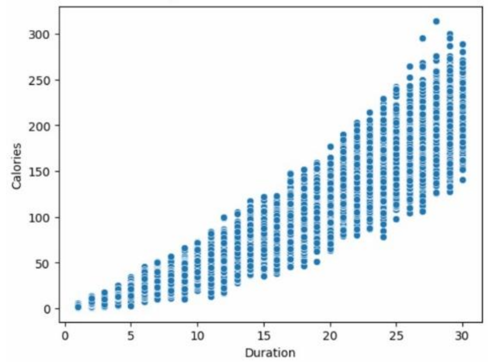


Heat maps are a type of color gradient in two-dimensional tables used to represent numbers. They are suitable for displaying high-volume data due to their dense and intuitive display method. They are frequently used to illustrate multivariable data because of their ability to display hundreds of rows and columns on a screen. They primarily rely on color intensity.[10]

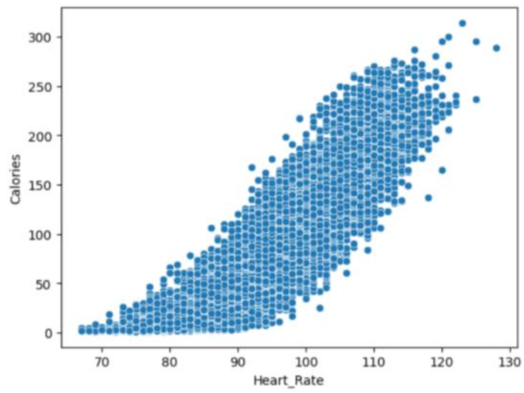
In the heat map, we see the correlation between each numerical characteristic. We see a very strong correlation between calories burned and heart rate, body temperature, and exercise duration. The strongest relationship is between exercise duration and the calories burned with correlation of (1). This means that the longer the exercise duration, the more calories burned. The second strongest relationship between calories burned is with heart rate., with a correlation of (0.9). The higher the heart rate, the higher the rate of calorie burning. Next is body temperature, with a correlation of (0.8). Similarly, the higher the body temperature, the higher the rate of calorie burning. However, there is no relationship between the target variable (calories) and height or weight and also the age because it has very small correlation.

Scatter plots between target and key features:

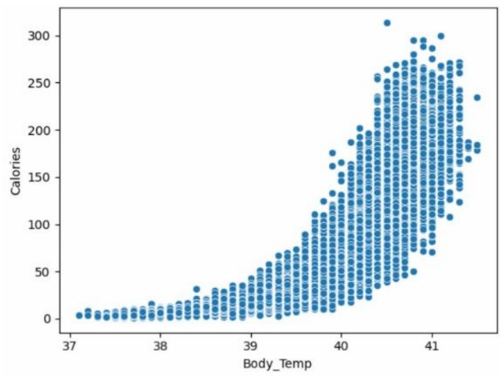
Scatter plots are plots used to display and visualize two-dimensional data as points and that is done in a simple and easy way. These plots help us identify trends and correlations between data, easily identify outliers, and compare different data sets by plotting them on the same axes. Therefore, scatter plots are useful and important for data exploration.[11]



According to the scatter plot above, we can see that the x-axis represents the duration of the exercise, while the y-axis represents the target variable, which is the number of calories burned. From the diagram, we can observe a representation, in the form of dots, of individuals based on their exercise duration and calorie burn. There is clearly a positive trend between calories burned and duration of the exercise; calories burned increase with increasing activity duration at a consistent pace. This indicates that those who want to burn more calories should increase their exercise duration, while those who want to maintain a lower calorie burn should reduce their exercise duration.



The scatter plot clearly illustrates a positive trend between calories (target variable) and heart rate. According to the scatter plot above, we can observe that the heart rate ranges from 70 (the normal heart rate) to over 120. The relationship can be explained as follows: the calories consumed (y-axis) increase with the rise in heart rate (x-axis). This is because during exertion or exercise, the heart rate increases, leading to more effective calories burning. Therefore, monitoring heart rate is important during exercise to determine the number of beats needed to burn a certain number of calories.



According to the scatter plot above, we can see that the x-axis represents body temperature, while the y-axis represents the target variable, which is the number of calories burned. The graph shows a dotted representation of individuals based on their body temperature and calories burned. Body temperatures range from 37°C to over 41°C, while calorie expenditure ranges from zero to over 300 calories. We observe a positive correlation between calories burned and body temperature. Therefore, we can understand that an increase in body temperature leads to higher calorie expenditure, and thus, we can increase calorie burn by monitoring body temperature until it rises to the required amount.

5) **Drop unnecessary column:**
When using a large dataset, it is important to simplify the solution by removing unnecessary

columns that do not affect the analysis. In the Pandas library that we used, we can delete one or more columns from the data frame using (df.drop['User_ID'], axis=1).[12]
We removed the ID because it is useless in the process of data analysis and model training.

6) **Encode categorical variables:**
Feature Encoding converts categorical data into numerical form so that machine learning models can process it efficiently. Many ML algorithms cannot work with raw categorical data, so encoding is necessary.

We stored the label encoder in a label variable and then used label fit transform to encode the data and convert the gender column to a numeric column (0 => female, 1 => male).

7) **Split dataset into training (70%) and testing (30%) sets:**
First, we split the dataset into X and Y. We gave the X all the columns except the calories, where X represent the input and we gave Y the calories, which is our target. We divided the test into 70% training and 30% testing.

8) **Normalize or scale numerical data:**
We use Standard Scaler to scale the data. We stored the standard scaler in the variable scaler. We used scaler fit and transform on the numeric X_train data and only used transform on the test data (Note: We did not use fit on the test data because we do not want to give prior knowledge of the test data).

C. *Proposed Method*

In our project, we started the project with data collection then exploration and understanding the data and other Dara Preprocessing tasks. we compared two algorithms to find the best one that help us build a successful model for calculating calories burned more effectively. We built a model, trained it, and tested it on new data.

Model Building:

Understanding the database and its nature is essential for knowing and selecting the appropriate algorithms to create a predictive model. To choose the right algorithm, it is important to understand each algorithm and its interpretability [13].

We built a model using two algorithms: linear regression and decision tree, to predict calories burned.

First, we must understand what decision trees and linear regression algorithms are. Decision trees and linear regression are important aspects of data analysis and prediction. [14]

Linear regression: This is a statistical method used to create a model by fitting data in a straight line to model the relationship between a dependent variable (Y) and one or more independent variables (X). Linear regression algorithms are easy to interpret the coefficients and are also effective for large datasets. [14]

Decision trees: These are a type of machine learning and data mining algorithm. Decision trees use decisions to build a tree-like model based on input data. Each internal node represents a decision, and this decision is based on a specific attribute. Therefore, we have different branches, and the leaf node represents the outcome or prediction. Decision trees are easy to interpret due to their tree-like structure. [14]

We imported “LinearRegression” from “sklearn.linear_model” and stored it in a “reg” variable. We then imported the “Decision Tree Regressor” from the “sklearn.tree” library.

Model Training:

We can now effectively train the model after building it by using the data designated to teach the model how to recognize patterns and make predictions. We use the algorithm we selected to input the training data, and the algorithm identifies patterns to minimize the difference between its predictions and the target values in the training data. This makes the model more skilled at making accurate predictions.[13]

We then trained the model on 70% (X_train, Y_train) of the data to improve prediction accuracy by using the `fit` method on the model variable.

Model Testing:

Experimental studies show us that if we use 20-30% of the available data for testing, we can achieve the best results.[15]

We tested the model using the test data, which represents 30% of the total data. We stored the Y-values predicted by the model in the linear regression variable (lr_pred) and the decision tree variable (dt_pred).

III. RESULTS DISCUSSION AND LIMITATION

A. *Algorithms results*

Model Evaluation
After a Model Training, you now need to evaluate its performance because it may not be perfect due to many factors that affect it, such as measurement errors. Therefore, there are different metrics we use to assess the model's performance. Regression data contains some common functions.[13]

We imported these three functions from the “sklearn.metrics library”: “mean_absolute_error”, “mean_squared_error”, and “r2_score”.

Mean absolute error:

Measures the average of the distance between correct and expected values and gives equal weight to all errors.

To calculate the mean absolute error in Python, we sum the absolute difference between the actual and calculated values for each observation across the entire array, then divide the result by the number of observations in the array.[16]

MAE=

$$\frac{1}{n}\sum_{i=1}^{n} | y_i - y\hat{}_i |$$

We can do this using the mean_absolute_error() function, which we call from the sklearn.metrics library in Python. This function performs machine learning models that simplify the calculation of the mean absolute error by handling all the necessary steps internally, ensuring accuracy and efficiency, especially with large datasets. [16]

These are the results we obtained from calculating the Mean Absolute Error.

Linear regression: 8.204985730307847

Decision tree: 8.77313249652086

A low MEA value of (8.204985730307847) for linear regression indicates a higher prediction accuracy. The lower the average, the higher the accuracy, and the higher the average, the lower the accuracy. As we observe on the slightly higher MEA value for the decision tree of (8.77313249652086), its accuracy will be lower and therefore less efficient in prediction. This indicates that the efficiency and accuracy of linear regression in prediction is higher, and we can rely on it in our model for satisfactory results.

Mean squared error:
The mean squared error (MSE) is a crucial concept in statistics and machine learning, playing a vital role in evaluating the accuracy of predictive models. The MSE value helps us analyze model accuracy by measuring the squared difference between the predicted and actual values in a data set. It is used in diverse fields such as statistics and machine learning. This is because it assigns greater weight to larger errors than smaller ones, making it sensitive to outliers. It provides a quantitative measure of the accuracy of predictive models and is easily interpreted mathematically. [17]

The equation used to calculate the mean squared error is as follows:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

Where: **n** (number of observations), **yi** (the actual value of observation), $\hat{Y}_i$ (the predicted value of the i^{th} observation). [17]

However, we can implement and calculate the mean squared error by calling the mean_squared_error from the sklearn.metrics library. Using this function, we can calculate the mean squared error in an easy and accurate way that does not take much time and is effective with data of different sizes.

These are the results we obtained from calculating the Mean squared error.

Linear regression: 122.97240835750641

Decision tree: 154.26691590820826

From the values above, we can observe that the MSE (122.97240835750641) for the linear regression model is lower than the MSE (154.26691590820826) for the decision tree model. In the MSE scale, lower values indicate higher model accuracy. Therefore, we conclude that a linear regression model with a lower MSE has higher accuracy and, consequently, better performance. Conversely, a decision tree model with a higher MSE has lower accuracy and lower performance.

R^2 score:
The **R^2 score** (coefficient of determination) is the statistical measure in regression analysis that deals with independent and dependent variables, meaning that a change in the independent variable leads to a change in the dependent variable. Essentially, the R^2 coefficient measures how well the regression line fits and predicts, or in other words, it measures the extent to which the independent data can explain the dependent data.[18]

$R^2 = 1 - (RSS/TSS)$
Where,

- R2 represents the required R Squared value,
- RSS represents the residual sum of squares, and
- TSS represents the total sum of squares.

We can do this using the r2_score () function, which we call from the sklearn. metrics library in Python. This function executes machine learning models that simplify the calculation of the R^2 score by handling all the necessary steps internally, ensuring accuracy and efficiency, especially with large datasets.

These are the results we obtained from calculating the R^2.

Linear regression: 0.9677395976320939

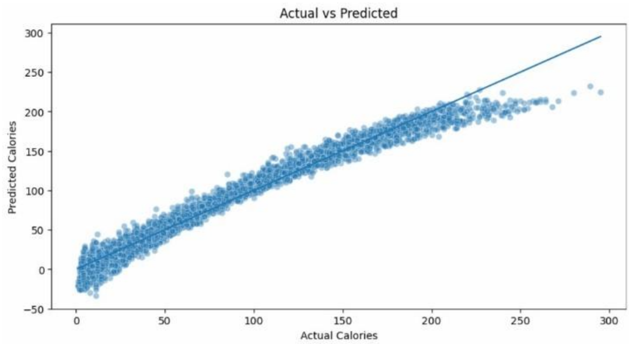
Decision tree: 0.9595298421351041

The higher **R^2 score** value for the linear regression (0.9677395976320939) indicates more accurate predictions; the higher the **R^2 score**, the higher the accuracy. Meanwhile, the slightly lower **R^2 score** value (0.9595298421351041) for decision trees suggests that linear regression is less accurate; the lower the **R^2 score** value, the lower the accuracy, which suggests that linear regression is more accurate.

Actual vs Predicted Plot:

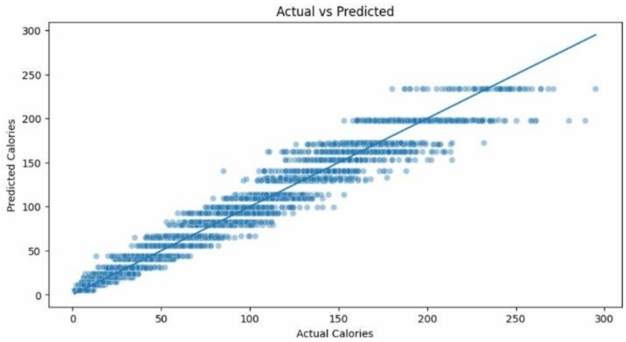
Plotting actual and predicted values is a way to visualize the difference between the actual and predicted values produced by the model. We used two graphs to determine the Actual and predicted values: for Linear Regression and Decision Tree. This graph helps us evaluate the model's performance and determine how close the result is to the correct values. To plot the predicted and actual values, we first need to collect data for both the predicted and actual values. Second, we distribute this data on the graph so that the x-axis represents the actual values and the y-axis represents the predicted values. Then, we compare them to see if the predicted values give accurate results that match the actual values. We can determine this by observing if the relationship is linear and the points lie on the same drawn line. Conversely, if the points are scattered and irregular throughout the graph, this indicates a lack of accuracy in the predicted values, and the model is not functioning correctly. [19]

Actual vs Predicted for Linear regression:



According to the line plot for linear regression, we placed the actual calories burned on the x-axis and the predicted calories burned on the y-axis. The blue dots represent the intersection of the predicted and actual calorie values. We drew a dividing line to highlight the linear relationship of the model. The results were remarkable. The graph indicates a linear relationship and that the prediction was highly accurate and satisfactory. Initially, the prediction was more accurate than its final stage, but this doesn't mean the final stage was poor. Overall, the model's prediction was very good. The calorie prediction was very close to the actual calorie expenditure, demonstrating that it is a good predictive model.

Actual vs Predicted for Decision tree:



According to the line plot for the decision tree, we plotted the actual calories burned on the x-axis and the predicted calories on the y-axis. The blue dots represent the intersection of the predicted and actual calorie values. We drew a line break to illustrate the linear relationship of the model. The results were fairly good, but they are not comparable at all to the results of linear regression. In the graph here, we can see that the distribution is not very uniform and that there are scattered points. This indicates lower accuracy and therefore a low-efficiency prediction, which we cannot rely on in our model because it has many prediction errors.

Overall, the more accurate results of linear regression indicate that it is the most suitable algorithm for calculating burned calories compared to the decision tree. This is because after evaluating the model's performance using three metrics (Mean Absolute Error, Mean Squared Error, R^2 score), we found that in all three measures, the results of linear regression were more accurate than those of the decision tree. We confirmed this through the graph of actual values versus predicted values for both linear regression and the decision tree. For linear regression, the results were excellent. The graph clearly showed a linear relationship, and the predictions were highly accurate and satisfactory. As for the decision tree, the results were somewhat good, but they cannot be compared to the results of linear regression. Therefore, we relied on linear regression in our model.

B. Limitation

In designing the model, we didn't encounter any major problems, but the fact that we didn't compare all available algorithms might pose a limitation.

IV. Conclusion

In conclusion, we would like to summarize some of the points mentioned. We discussed obesity and its recent spread, and the importance of our model in helping to solve this problem. We tried several algorithms to obtain better results, and ultimately found that the linear regression model was the most suitable. For future work, you can try training other algorithms that help in training the calorie-burn prediction model and provide higher accuracy. As a piece of advice, we remind you of the importance of maintaining your physical health and being aware of your calorie intake to live a healthy life. Our model will facilitate your healthy lifestyle.

REFERENCES

[1] The General Authority for Statistics (GASTAT), GASTAT publishes the results of Health Determinants Statistics Publication in Saudi Arabia.2024.[Online]. Available: The link of the website

[2]LUMENALTA,"13 benefits and challenges of machine learning".2024 .[Online]. Available: The link of the website

[3] McKinney, W. (2011). pandas: a foundational Python library for data analysis and statistics. *Python for high performance and scientific computing*, *14*(9), 1-9. The link of the website

[4] Sapre, A., & Vartak, S. (2020). Scientific computing and data analysis using NumPy and Pandas. *International Research Journal of Engineering and Technology*, *7*, 1334-1346.
The link to the research paper

[5] Idreos, S., Papaemmanouil, O., & Chaudhuri, S. (2015, May). Overview of data exploration techniques. In *Proceedings of the 2015 ACM SIGMOD international conference on management of data* (pp. 277-281). Available: The link of the website

[6] Ray, S., Alshouiliy, K., & Agrawal, D. P. (2020). Dimensionality reduction for human activity recognition using google colab. *Information*, *12*(1), 6. Available: The link to the research paper

[7]Laerd,Measures of Central Tendency,n.d,[Online].Available: The link of the website

[8] LibreTexts, "Basic statistics- mean, median, average, standard deviation, z-scores, and p-value". 2025.[Online].Available: The link of the website

[9]GeeksforGeeks.Histogram-Definition,Types, Graph,and Examples .23 Jul, 2025.[Online].Available: The link of the website

[10] Gehlenborg, N., & Wong, B. (2012). Heat maps. *Nature Methods*, *9*(3), 213. The link to the research paper

[11] Mayorga, A., & Gleicher, M. (2013). Splatterplots: Overcoming overdraw in scatter plots. *IEEE transactions on visualization and computer graphics*, *19*(9), 1526-1538. The link to the research paper

[12] GeeksforGeeks.How to Drop One or Multiple Columns in Pandas DataFrame .01 Nov, 2025. [Online].Available: The link of the website

[13] GeeksforGeeks. Steps to Build a Machine Learning Model . 2025.[Online]. Available: The link of the website

[14] GeeksforGeeks. Decision Trees vs Clustering Algorithms vs Linear Regression. 23 Jul, 2025 .[Online]. Available: Link of the website

[15] Why 70/30 or 80/20 Relation Between Training and Testing Sets: A Pedagogical Explanation. The link to the research paper

[16] GeeksforGeeks. How to Calculate Mean Absolut ErrorinPython?.27May,2025.[Online]. Available: The link of the website

[17] GeeksforGeeks. Mean Squared Error. 16 Sep, 2025.[Online]. Available: The link of the website

[18] GeeksforGeeks. R Squared | Coefficient of Determination . 23 Jul, 2025.[Online]. Available: The link of the website

[19] ResearchGate. Actual and predicted values plot. May 2023. https://www.researchgate.net/figure/Actual-and-predicted-values-plot-is-a-visualization-technique-used-to-compare-the-actual_fig1_370547323